

EDFA: Ensemble deep CNN for assessing student's cognitive state in adaptive online learning environments

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ABSTRACT

Ensuring student engagement is crucial for effective learning outcomes in any classroom setting, including e-learning environments. However, the absence of immediate supervision in online classes makes monitoring and maintaining student attentiveness difficult. To address this challenge, this study proposes a cognitive state detection system that continuously monitors the facial emotion of the learner in an adaptive learning environment. The algorithm is proposed to detect cognitive states such as attentiveness and inattentiveness. The system has been implemented on four separate databases and evaluated using three ensemble models: FT-EDFA, FC-EDFA, and OT-EDFA. The ensemble models have been created by applying transfer learning to two popular pre-trained models, VGG19 and ResNet50, which can learn useful features from facial images for emotion recognition tasks. Combining the features learned by both models, the ensemble approach can achieve better performance in recognising facial emotions. The proposed system can provide continuous feedback to instructors, enabling them to adjust their teaching methods to maintain student engagement and interest. The study has achieved promising results, surpassing the performance of existing methods with recognition rates of 93.11%, 92.34%, and 91.12% on the newly created dataset. By detecting cognitive states in online learners, the proposed system can help instructors understand how engaged and interested their students are during classes. Overall, facial emotion recognition can be useful for improving the quality of e-learning platforms and enhancing student learning outcomes.

1. Introduction

In e-learning environments, the ability to capture a learner's emotions and adjust the course material is often lacking (Mishra et al., 2020). When students become disengaged or bored during important lectures or courses, it can nullify the efforts of the instructor and their instructional aides, resulting in a less effective learning experience (Kundu & Bej, 2021). Meanwhile, to tackle this problem, various interactive learning technologies have been introduced recently (Aguilera-Hermida, 2020; Goar et al., 2021; Omura et al., 2020). Therefore, there is a pressing need for a system that can effectively analyse a student's cognitive state in real-time within an e-learning environment (Adedoyin & Soykan, 2020; Gupta et al., 2023b; Kaur et al., 2021). The term *cognitive state* is used here to describe the student's mental disposition while studying. Detecting the cognitive state of the learner is important in an adaptive learning context to know whether the learner is engaged or not. If the learner is not engaged, proper feedback is immediately provided; otherwise, learning performance will decrease. It is important

to incorporate elements of physical classrooms into online learning to create a well-rounded educational experience for online learners. Using facial expressions to convey emotional reactions has been a subject of scientific interest for over 150 years (Zheng et al., 2021; Gupta et al., 2023a). Analysing facial expressions by extracting important facial features is a popular way to describe facial emotions (Ekman & Oster, 1979; Kaur et al., 2023). It can tell the difference between the attentive or inattentive cognitive state of the learner. The demand for more advanced human-machine interfaces has driven computer vision's attention to automatic emotion identification from facial expressions (Mukhopadhyay et al., 2020; Sharma et al., 2019).

However, automatic recognition of emotion in real-time scenarios is challenging, requiring sophisticated algorithms to accurately identify and interpret facial expressions. Despite these challenges, advancements in computer vision technology have enabled the development of more efficient and effective Facial Emotion Recognition (FER) systems. By accurately detecting and interpreting a learner's expressions, such a system can adjust the course material dynamically to elevate their

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learning performance and maintain engagement. This would enable instructors to design more effective and personalised learning experiences tailored to each student's preferences and requirements. Our results indicated a strong correlation between a learner's emotions and their level of interest or involvement during online lectures, which can be used to vary the content to improve their engagement in the online learning system. Overall, our study demonstrates the potential of utilising facial expression recognition to enhance the effectiveness of adaptive learning platforms and promote better learning outcomes and academic success.

So, the objective of the suggested system is to develop an efficient Real-Time Cognitive state Detection System that can be used in smart and interactive adaptive learning environments. The proposed study's contributions are highlighted below:

- An Ensemble-based cognitive state detection system is proposed that recognises facial emotion in the adaptive learning context.
- Facial emotion data is collected using the web camera, and data is pre-processed to clean it and make it consistent and meaningful for analysis.
- Face region is identified in a live video stream using the Haar-cascade algorithm.
- Feature extraction is done on the pre-processed data to get the important features for further processing.
- We present an ensemble-based approach for classifying distinct emotions based on the extracted features of the face, i.e., by analysing facial expressions.
- Attention index (AI) is calculated from the proposed algorithm to predict the cognitive state (either attentive or inattentive) during the online learning session.
- To provide the user with feedback in real-time in order to improve their learning and increase their productivity.

The study is structured as follows. Section 2 provides a brief overview of previous research on detecting cognitive states in adaptive learning environments. In Section 3, the datasets used in the study are discussed, while Section 4 covers the proposed techniques and system. The proposed algorithm is presented in Section 5. Experimental findings are presented in Section 6, which includes the results of experiments conducted to evaluate the proposed approach. Section 7 provides an analysis of the proposed model. A comparison of the proposed system with recent models is provided in Section 8. The study concludes in Section 9 by discussing future research directions.

2. Related work

Emotions play a crucial role in learning (D'Mello, 2013, Shao et al., 2019). They can affect learning in both positive and negative ways (Fiedler & Beier, 2014, Pardos et al., 2014, D'Mello et al., 2011) and can be leveraged to enhance the learning experience (Giannakos et al., 2021). With the emergence of internet-based teaching and e-learning techniques, the quantity and quality of education have dramatically increased. However, a teacher's lack of direct supervision in digital learning environments can hinder a student's ability to learn effectively.

During lectures, educators can gain insights into the satisfaction levels of their students (Scagnoli et al., 2019), which can impact the level of engagement that students have in the class, which is closely linked to the instructor's effectiveness (Franklin & Harrington, 2019). Direct supervision is crucial for facilitating learning and keeping students aligned with the course objectives, and this is possible through immediate communication with the instructor during the lecture. High frustration levels result from a lack of communication (Camacho-Zuñiga et al., 2021, Yadav et al., 2021, Singh et al., 2022). Although supervised teaching is crucial for learner's academic progress, virtual learning creates distinct hurdles for educators and learners; since remote students might not have access to a physical school and may encounter obstacles in building relationships with fellow students and faculty members. Due

to the absence of immediate communication channels, e-learning platforms solely rely on standard man-machine interfaces to comprehend recorded data (Mukhtar et al., 2020). As there is no direct dialogue between an instructor and a student in an online setting, facial expressions are considered key indicators of a student's emotional state and involvement during class.

In (Ekman et al., 2013) have conducted extensive research on the relationship between facial features and emotions, which is considered a major contribution to the analysis of emotions based on facial attributes (Ko, 2018, Kang et al., 2019). Even though the facial action coding system can capture immediate emotional responses on a person's face, it is essential to consider different action units to determine a complete mood, as they may vary based on an individual's personality and circumstances. The technology of recognising faces is based on identifying facial attributes like the mouth, forehead, nose and eyes, which help detect expressions on human faces (Kumar et al., 2019). This technology can quickly detect and classify facial expressions in real-time, and the accuracy and detection time can be improved using machine learning algorithms (Hussain & Al Balushi, 2020, Deepa et al., 2021). Facial expressions are the result of emotional impulses that trigger physical muscle movements, such as curling of lips, raising eyebrows, and wrinkling the forehead. The use of intelligent methods to extract facial expressions for faster image analysis has been suggested by researchers (Kumari et al., 2015). Various studies have populated databases of facial expressions, leading to the development of intriguing algorithms for various applications. Various techniques have been explored to detect emotions; however, none of them has shown significant accuracy in creating a practical system for detecting a person's attention while learning. A few deep CNN-based techniques that have been used are VGG19 and ResNet50 (Maamari & Majdalani, 2019, Sanchez-Ruiz et al., 2021, Thomas et al., 2021). Deep CNN is known for its high-speed computation and has been applied in image retrieval and face recognition (Gonçalves et al., 2021, Sharma et al., 2021, Shad et al., 2021). In one study, features of VGG and ResNet were combined to extract different facial features. VGG was employed to classify the extracted features; however, the accuracy was relatively low due to the limited amount of data and a lack of variation in the images. One approach that researchers have attempted to improve accuracy in attention detection is through the use of ensemble neural networks. A multi-layered neural network with five layers was implemented, but its accuracy was less than 60% (Verma & Khunteta, 2017, Prakash & Karam, 2021). Another method that has been explored is the VGG16 model, which is an extension of the convolutional neural network (CNN). VGG16 uses more layers than CNN (Sarkar et al., 2022, Kumar et al., 2022, Praveen et al., 2021) and employs fully connected and max pooling layers for emotion classification, but it still achieved an accuracy of only 65% (Prakash et al., 2019, Arafah et al., 2019). Despite advancements in research and implementation in analysing a user's mood, this technology is seldom utilised in practical applications. This may be due to the system's inefficiency or lack of accuracy, or the technology's availability may not be widespread. To address the major issue of the absence of methods to recognise the behaviour of learners in distance learning, it is crucial to have effective interactive models in place. The present study suggests one possible solution to develop a cognitive state detection system that assesses the emotional aspect of e-learners while using adaptive learning platforms.

3. Data sources

To evaluate the efficacy of our system, we utilised three distinct publicly available databases. These databases include the CK+ database, an extended version of the Cohen-Kanade dataset, the FER-2013 dataset for facial expression recognition, and the RAF-DB dataset containing real-world affective faces. One more dataset has been created in this study, and the proposed approach is evaluated on this newly created dataset. The description of each dataset is given below.

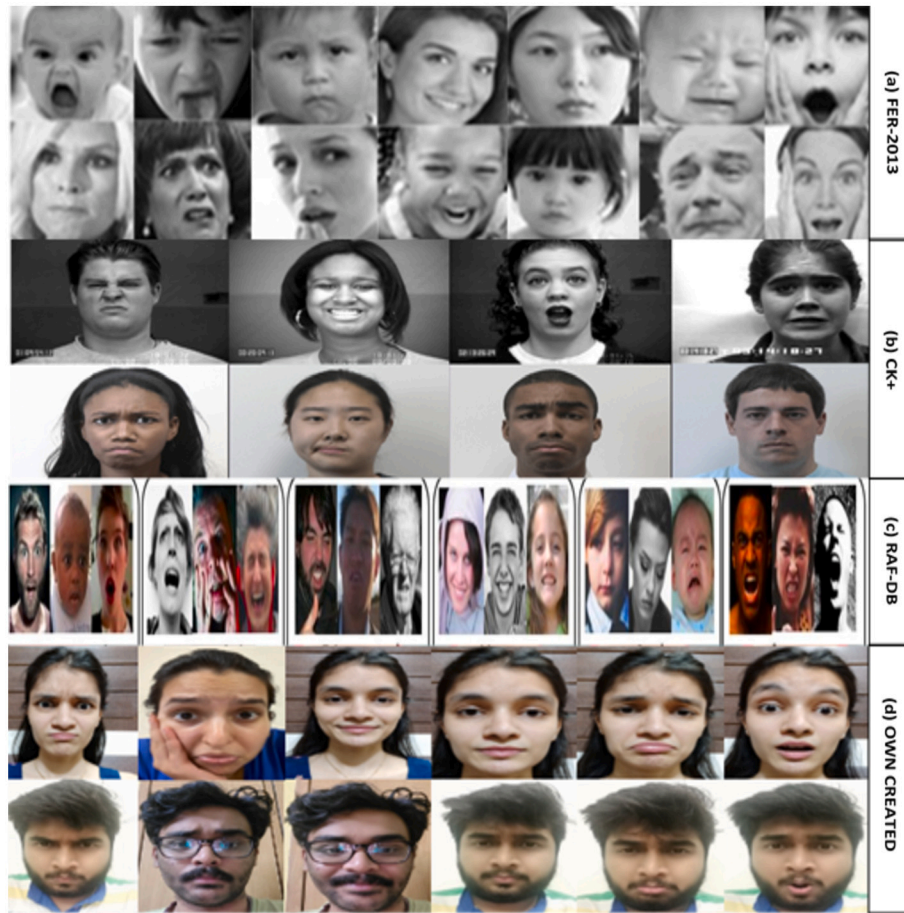


Fig. 1. Examples images from (a) FER-2013 (b) CK+ (c) RAF-DB and (d) own datasets.

Table 1

Summary of FER-2013 dataset.

NAME	YEAR	NUMBER	COLOUR	SIZE	Different EXPRESSIONS
FER-2013	2013	36,000	Gray	48X48	Neutral, Happy, Fear, Disgust, Contempt, Angry, Sad, Surprise

3.1. Facial Expression Recognition 2013 (FER-2013)

The ICML's Kaggle facial expression recognition challenge in 2013 featured a popular dataset called FER-2013 (Facial Expression Recognition 2013), which has since become widely used (Giannopoulos et al., 2018). Detailed information about the FER-2013 dataset can be found in Table 1, while sample images from this dataset are displayed in Fig. 1(a).

3.2. Extended Cohn-Kanade dataset (CK+)

Researchers commonly rely on the Extended Cohn-Kanade (CK+) (Gupta, 2018) image database to test and validate their FER methods, whether in static or dynamic (video) settings (Malik et al., 2021). The CK+ dataset boasts 594 sequences of video featuring 123 individuals aged 18 to 50. The images in the CK+ database were taken from frontal angles and had resolutions of either 640×490 or 640×480 pixels. While the CK+ database includes colour images, they were converted to grayscale for this study. Fig. 1(b) displayed sample images from this dataset, and its detailed description showed in Table 2.

Table 2

Summary of CK+ dataset.

NAME	YEAR	NUMBER	COLOUR	SIZE	Different EXPRESSIONS
CK+	2011	594	Gray/Colour	640X480	Neutral, Happy, Fear, Disgust, Contempt, Angry, Sad, Surprise

Table 3

Summary of RAF-DB dataset.

NAME	YEAR	NUMBER	COLOUR	SIZE	Different EXPRESSIONS
RAF-DB	2017	30,000	Colour	350x350	Neutral, Happy, Fear, Disgust, Contempt, Angry, Sad, Contempt

3.3. Real-world affective faces (RAF-DB)

With over 30,000 facial images have been annotated by 40 human coders trained in recognising basic or compound expressions, the RAF-DB (Ryerson Audio-Visual Database of Emotional Speech and Song) is a valuable resource for researchers (Greco et al., 2022). However, in our particular experiment, we only used images representing basic emotions. We used 12,271 photos for training and 3,068 for testing, both taken from this subset of the RAF-DB. Fig. 1(c) displayed sample images from this dataset, and its detailed description is given in Table 3.

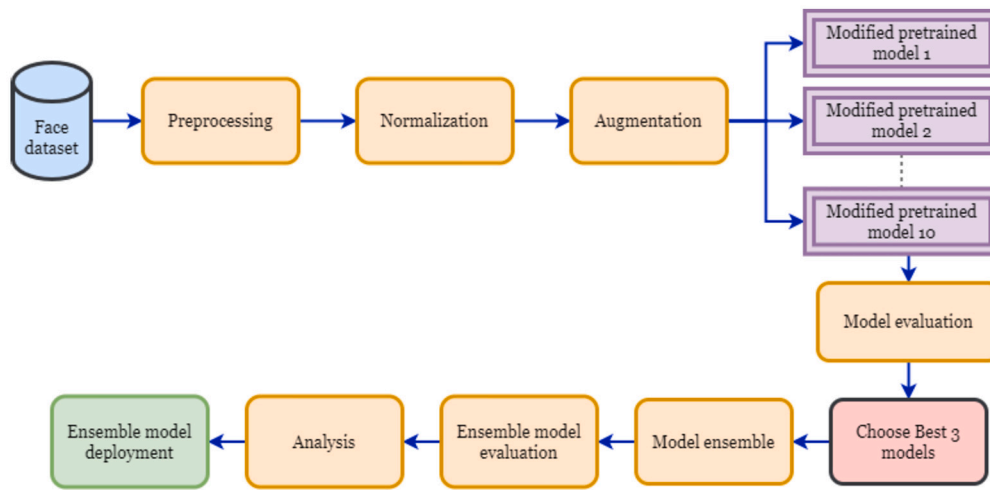


Fig. 2. Proposed Flowchart of cognitive state detection system design.

Table 4
Summary of own dataset.

NAME	NUMBER	COLOUR	SIZE	Different EXPRESSIONS
OWN collected Dataset	1950	Colour	48 X 48	Neutral, Happy, Fear, Surprise, Angry, Sad

3.4. Own dataset

To classify facial emotions, a novel dataset was created by gathering pictures of 200 undergraduate students. Each photo was taken with depth and RGB in mind, and subsequently, each individual was assigned labels for six basic emotions. Fig. 1(d) showcases some sample images from the dataset, and an overview of the dataset is provided in Table 4.

4. Methodology for assessing cognitive state

This section is dedicated to the development and implementation of the ensemble model for detecting cognitive states. The main focus of this study is to differentiate between attentive and inattentive states. The steps followed are shown in Fig. 2. It provides a visual representation of the key steps involved in the process. In the first stage, from the live video streaming, images are captured by the in-built camera. The facial region is extracted from the captured image using the Haar-cascade algorithm. Next is pre-processing and feature extraction of the data obtained. This is necessary because different images may have different scales, which can result in inaccurate detection of facial points. During this step, facial features such as eyebrows or mouth corners are measured to obtain more meaningful data. Once the data has been pre-processed and features have been extracted, the model is trained using an ensemble deep neural network. Deep CNNs are a popular deep-learning algorithm for recognising and classifying images (Druzhkov & Kustikova, 2016). The pre-processed data is fed into the proposed ensemble model, which consists of multiple layers of filters that learn to recognise patterns and features in the input data. The filter weights are iteratively modified during training to decrease the gap between the model's actual and predicted output. The number of epochs for training is determined during this process. The objective is to create a model that accurately classifies the user's cognitive state (classify them as either Attentive or Inattentive) based on the facial expressions captured during the video streaming. By analysing how each mood contributes to each parameter, we can make the system interactive and help users stay focused. Finally, we give real-time feedback to the user for effective and efficient learning. By alerting users to their Inattentiveness during

online learning, we can help them concentrate better and ultimately improve their learning outcomes. The steps followed are discussed below.

4.1. Pre-processing phase

The deep CNN models were trained using pixel values ranging from 0 to 255, which is the standard representation required in image processing. For this purpose, the equation EQ. (1) followed is given below:

$$Im_{scaled} = (Im - Mini) \frac{Maxi_{new} - Mini_{new}}{Maxi - Mini} + Mini_{new} \quad (1)$$

To ensure consistency, the pixel values were rescaled from the original range to a new range of 0 to 1 using a linear transformation equation. In this case, pixel values of 0 and 255 are represented by Min and Max, while Minnew and Maxnew represent pixel values of 0 and 1. To ensure uniformity across the training, validation, and testing phases, we rescaled all datasets' pixel values before using them. To optimise the image input for ensemble deep learning, the image frame was converted into grayscale and reshaped using NumPy. This reshaped image was fed into a pre-loaded Keras model using the "load model()" function, producing the maximum argument. During the execution phase, a combination of OpenCV and Keras functions was utilised.

In the present study, the video frame was initially saved into a video object; then, a Haar cascade classifier was used to identify facial regions within the image. The Haar Cascade method detects faces in video streaming. This method employs a set of rectangular filters called Haar-like features, which can detect patterns of intensity changes in the image. The features are applied at various scales and positions to detect faces of varying sizes and orientations. The Haar Cascade classifier is trained on a huge collection of positive and negative images examples to detect faces in different lighting conditions and backgrounds accurately (Fig. 3). Finally, a rectangular box was drawn around the facial regions, displaying the face of the user. Multiple models were trained and utilised for further processing, such as feature extraction and classification, to recognise the emotions expressed in the face.

4.2. Feature extraction

Feature extraction is a crucial phase in the image processing pipeline, as it aims to capture and represent salient information in the image that is relevant to the classification task. Recent advancements in CNN-based models have enabled the identification of emotional changes and recognising of learner's cognitive states in real time during online classes. Many deep CNN models are good for facial emotion recognition, but using a transfer learning model is more beneficial.

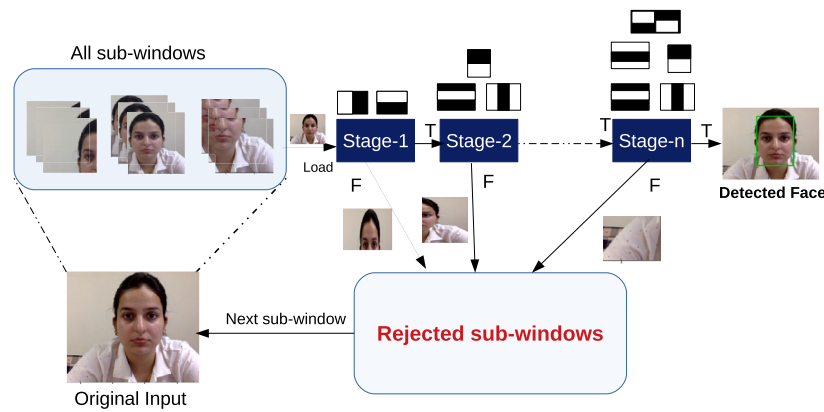


Fig. 3. Cascade structure for Haar classifiers to detect face in real-time video streams.

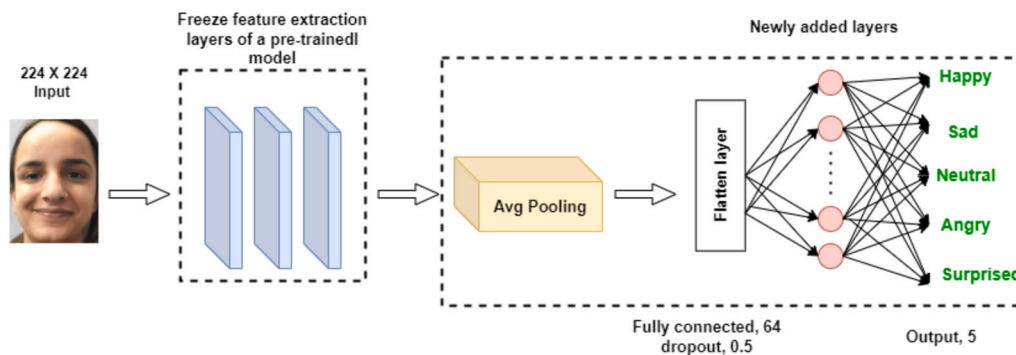


Fig. 4. Network Architecture of modified pre-trained model.

Training a CNN model from scratch can be time-consuming and computationally costly because the model needs a lot of data to learn useful features. Using transfer learning, we can take advantage of models that have already learned useful features from large datasets and then modify them for our specific task using a smaller data set. As a result, training can be completed more quickly and with better outcomes than when starting from scratch. Therefore, transfer learning enables us to perform well with less labelled data and time. Let's discuss below.

4.3. Transfer learning models

The lack of high-quality datasets makes it difficult to train deep CNN models from scratch to recognise emotional behaviour based on facial expression analysis (Kaya et al., 2017). We apply a library of pre-trained models and transfer learning techniques to solve this problem. Transfer learning's key benefit is that it allows data to be trained in less time and with fewer samples (Zhuang et al., 2020). A newly trained model can benefit from the experience of a previously trained model. We employed transfer learning for the Emotion recognition task by utilising two commonly used CNN models: VGG19 and ResNet50 (Gondkar et al., 2021). These models were chosen due to their effectiveness in computer vision. We used these models as our base to keep the optimised weights and froze the feature extraction layers, consisting of pooling and convolutional layer pairs. This method is useful for training models to detect cognitive states in the future since it minimises data loss and optimises feature extraction. The FER-2013, CK+, RAF-DB, and newly created datasets train the modified models. During each epoch, the models are validated against a separate validation set. As this study focuses on binary classification (attentive and inattentive classes), the output layer's number of neurons is set to two. The modified architecture of the two pre-trained models used in this study is depicted in Fig. 4.

This study presents an Ensemble approach for detecting the cognitive state of online learners. To achieve this, we evaluated the performance of four CNN-based models, namely VGG19, ResNet50, and Ensemble Learning. We briefly described each model's architecture in the below subsection.

–VGG19

VGG19 is a popular CNN model in the field of image analysis (Bansal et al., 2021). There are 19 layers, 16 of which are convolutional, 1 is a max-pooling layer, and 3 are fully connected. The max-pooling layer uses 2x2 size kernels with 2 pixels stride size to downsample the feature maps, whereas the convolutional layers utilise 3x3 size kernels with a 1-pixel stride size. Except for the final layer, which has just 1000 channels, the completely connected layers all have 4096. To avoid overfitting, dropout is used on the completely linked layers. Six distinct facial expressions may be identified and categorised using VGG19 and input photos of 48x48 pixels. Using backpropagation, the model is trained on a huge dataset of human faces annotated with labels indicating the subjects' emotional states. The trained VGG19 model can recognise facial expressions in real time, which is useful for applications such as adaptive learning environments. In this research, VGG19 is trained to identify and classify instances of six different expressions in input images of 48x48 pixels (Fig. 5).

–ResNet50

For facial emotion identification, ResNet50 is another favourite (Li & Lima, 2021). It employs residual connections to circumvent the vanishing gradient issue and has 50 layers (Figs. 6, 7). Fig. 8 depicts the framework of ResNet50. It's 5-stages deep, with convolution and identity blocks in each stage. There are three 1x1, 3x3, and 1x1 convolution layers in the convolution block's filters. The identity block comprises two convolution layers with 1x1 and 3x3 sized filters, respectively. The pooling layer has a 3x3 kernel size, while the convolution layer is 7x7. ResNet50 accepts input images that are 224 × 224 × 3. In this research,

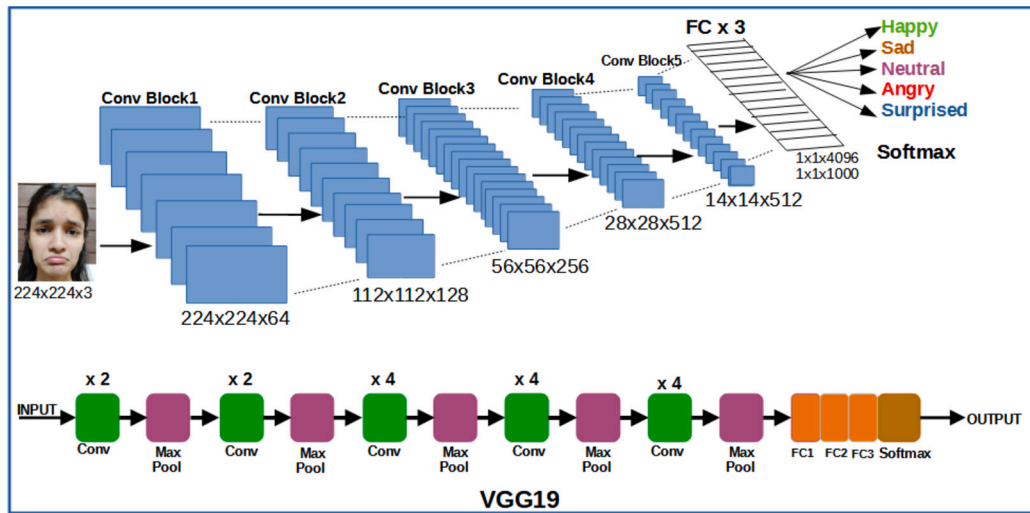


Fig. 5. Network Architecture of VGG19.

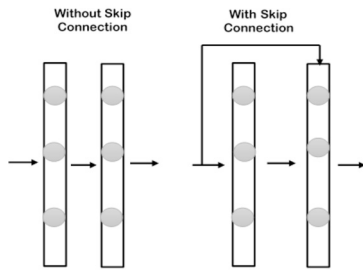


Fig. 6. Skip connection.

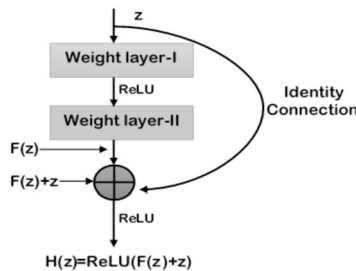


Fig. 7. Single Residual block.

224x224x3 input photos are utilised for training ResNet50 to classify facial expressions.

4.4. Ensemble learning

Our proposed method involves using an ensemble approach to combine two of the most successful deep learning models, VGG19 and ResNet50, to achieve improved accuracy. We extract feature vectors from each model's second to last layers and concatenate them to form a single feature vector with a dimension of 2560, as shown in Fig. 11. The facial emotion is classified based on the overall feature vector, training Ensemble model for each emotion. The model with the highest score determines the final prediction. We then calculate the AI (Attention index) based on the proposed Algorithm 1 to predict the cognitive state. By leveraging the strengths of the two models and generating an efficient ensemble model, our objective is to enhance emotion recognition results for our proposed system.

A large amount of variation may result from using the predictions of many deep learning classifiers separately because these classifiers were designed and trained differently. The weights are updated sepa-

ately, resulting in inconsistent results when classifying the same data. To address this, ensemble models can be used to decrease variance and improve generalizability. We present a deep neural network ensemble for emotion recognition by integrating two separate models. The feature extraction layers of the two best-performing pre-trained models were untrainable to maintain their optimised weights. Two different kinds of architecture are considered to build the efficient Ensemble learning model. Let's discuss below.

4.4.1. Feature-based EDFA (FT-EDFA)

The goal of the proposed FT-EDFA is to incorporate more features from the input into the model and its architecture is shown in Fig. 9. This is done with a technique called "feature ensemble," which combines all necessary features in the same face image. The FT-EDFA uses a combination of two of the most reliable pre-trained models available for identifying facial expressions.

Let's suppose Input (I) = $i_1, i_2, \dots, i_n$ as facial image. Then, this method can be displayed as:

$$WT_{FT} = \{wt_{FT1} + wt_{FT2} + \dots + wt_{FTn}\} \quad (2)$$

In the above equation EQ. (2), the value of $n=3$ and, $wt_{FT1} + wt_{FT2} + \dots + wt_{FTn}$ are the extracted features from the same input i_n by the transfer learning models. As a result, the feature ensemble that the several pre-trained models generated by the equation EQ. (3).

$$E_{FT} = \text{Concatenate}\{wt_{FT1} + wt_{FT2} + \dots + wt_{FTn}\} \quad (3)$$

Averaging pooling layers were added after the feature extraction layers, reducing the size of these two models. The results of these two individual feature extraction layers were then combined in an ensemble layer to get more accurate feature data. An output layer, a fully connected layer and a flattened layer have been included to conclude the feature ensemble model. The network Architecture of FT-EDFA

4.4.2. Fully connected-based EDFA (FC-EDFA)

FC-EDFA creates an ensemble model having 386 trainable parameters by fusing fully connected layers of two pre-trained models. The proposed model will get the input from the output of the trained model's fully connected layers as described in equation EQ. (4). Here, n equals 3.

$$WT_{FC} = \{wt_{FC1} + wt_{FC2} + \dots + wt_{FCn}\} \quad (4)$$

Therefore, below is a representation (given in equation EQ. (5)) of the ensemble of fully connected layers derived from different pre-trained models.

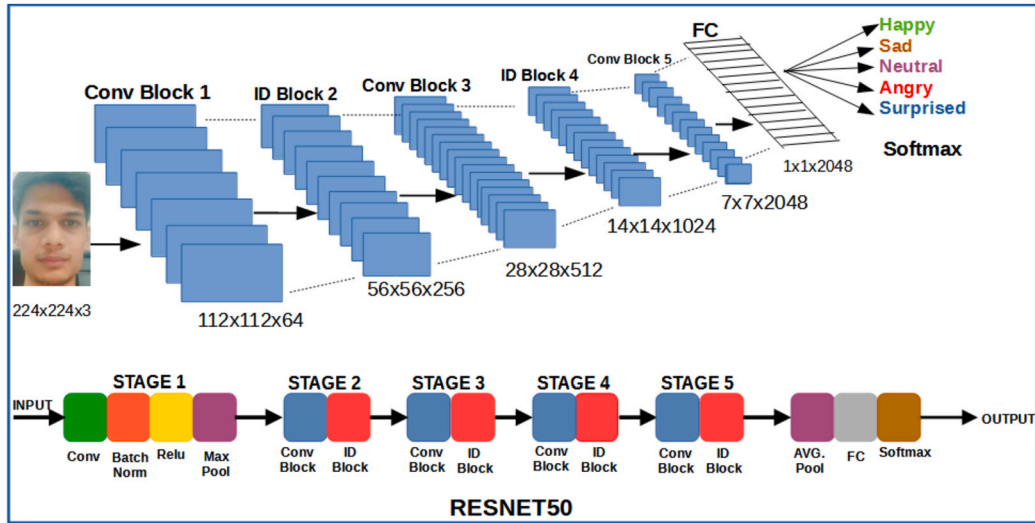


Fig. 8. Network Architecture of ResNet50.

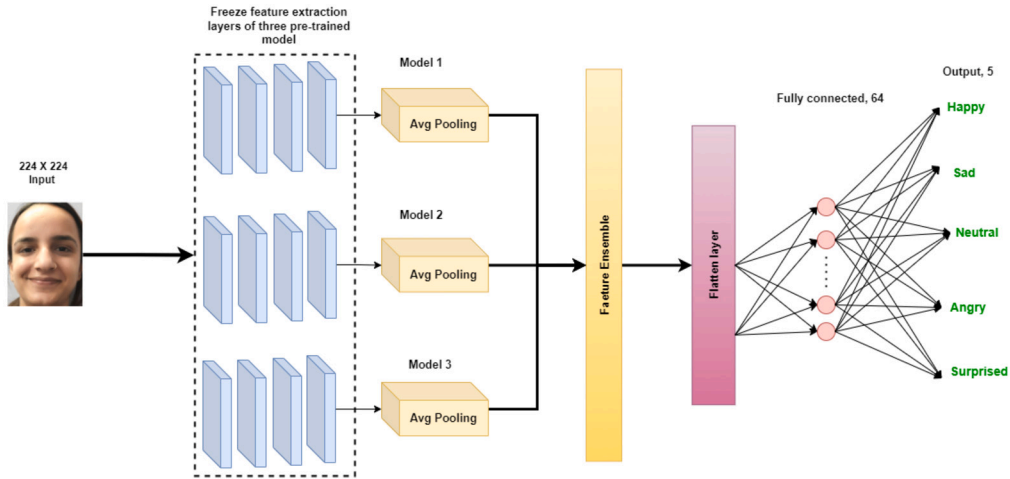


Fig. 9. Network Architecture of FT-EDFA.

$$E_{FC} = \text{Concatenate}\{wt_{FC1} + wt_{FC2} + \dots + wt_{FCn}\} \quad (5)$$

Combining the results of several fully connected layers improves the possibility of recognising facial emotion. The FC-EDFA model's architecture is depicted in Fig. 10.

4.4.3. Output-based EDFA (OT-EDFA)

The OT-EDFA strategy constitutes the assembly of two pre-trained models with 14 trainable parameters at the output layer. The equation EQ. (6) for this approach is as follows this:

$$WT_{OT} = \{wt_{OT1} + wt_{OT2} + \dots + wt_{OTn}\} \quad (6)$$

Here, $wt_{OT1}, wt_{OT2}, \dots, wt_{OTn}$ represent the results of separate pre-trained models. As a result, the output layer ensemble generated by the several pre-trained models can be expressed in equation EQ. (7) as follows:

$$E_{OT} = \text{Concatenate}\{wt_{OT1} + wt_{OT2} + \dots + wt_{OTn}\} \quad (7)$$

The proposed approach assumes that the merged output in the ensemble model can capture more essential characteristics and thus produce more accurate predictions. Fig. 11 illustrates the architecture of the O-EDNC model.

5. Ensemble learning based attention evaluation

This study presents a cognitive assessment system that uses the Ensemble model to detect the learner's cognitive state during adaptive learning. The cognitive state is calculated with input from an Ensemble neural network classifier trained to anticipate the learner's facial emotion. The Emotion Probability (EP) score and Emotion Weight, which represent the student's attention, are calculated by the Ensemble Neural Network classifier. As seen in Table 5, we have identified six distinct classes of facial expressions, and each of these groups has been further classified into two distinct categories: attentive and inattentive.

The predicted emotion is used as a factor in selecting the corresponding cognitive state. According to the EQ (8), the attention index is computed to identify the corresponding cognitive states.

$$EI = EP \times WE \quad (8)$$

Here, EP = Emotion Probability

WE = Weight value of Emotion

In the table provided, the relative value of each emotion is shown on a scale which is given in Table 6.

Following this, the EP score calculates the attention index, as described by the equation EQ. (8) in the proposed system. Below, we detail the Algorithm 1 used to determine one's attention via facial emotion analysis.

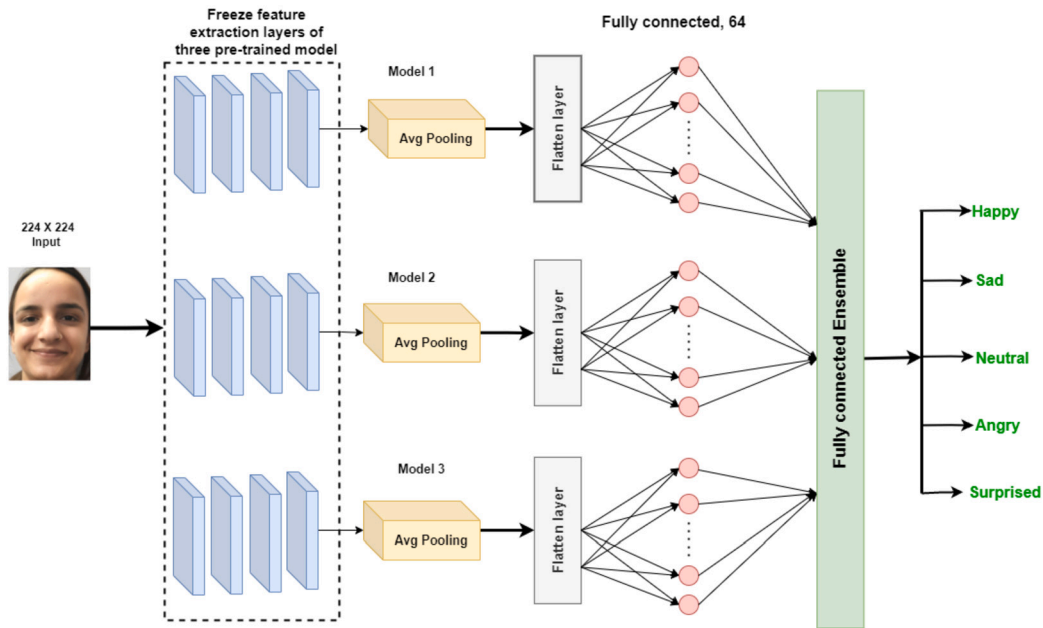


Fig. 10. Network Architecture of FC-EDFA.

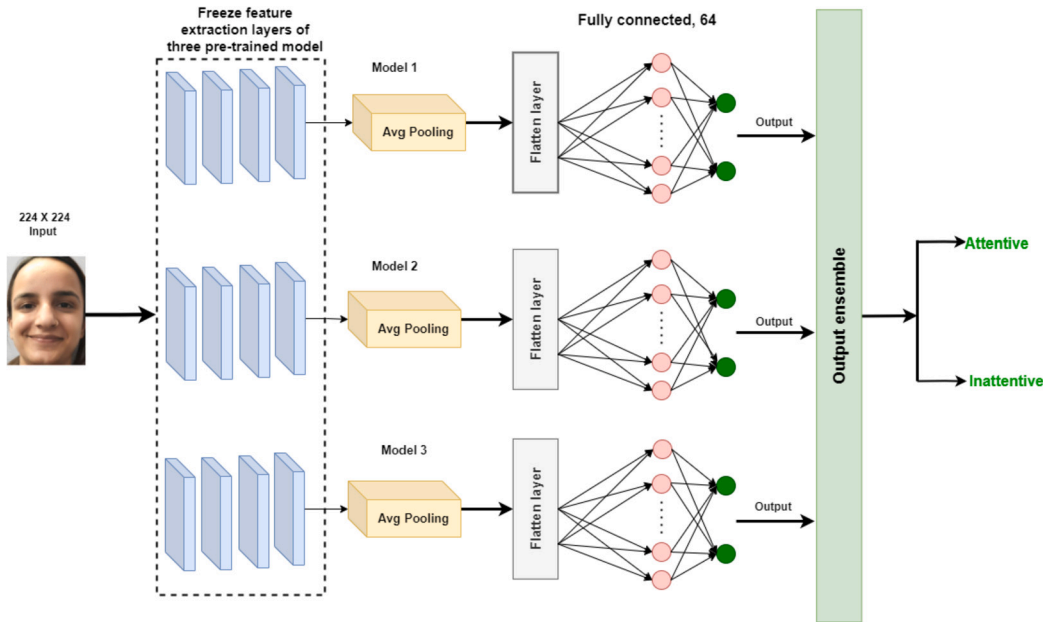


Fig. 11. Network Architecture of OT-EDFA.

Table 5

The Cognitive state determined by Algorithm 1's Emotional Pattern and EP Score.

Emotion	Cognitive state
Happy($EP > 60$)	Attentive
Surprised($EP > 70$)	Attentive
Neutral($EP > 65$)	Attentive
Angry($EP < 20$)+Neutral($EP > 65$)+Sad ($EP < 15$)	Attentive
Sad($EP > 20$)	InAttentive
Neutral($EP < 50$)	InAttentive
Fear($EP > 40$)	InAttentive
Angry($EP > 30$)	InAttentive
Surprised($EP < 50$)+Fear($EP > 40$)	InAttentive
Sad($EP > 25$)+Fear($EP > 25$)+Surprise($EP > 50$)	InAttentive
Fear($EP > 35$)+Sad($EP > 20$)	InAttentive
Sad($EP > 35$)+Angry($EP > 25$)	InAttentive

Table 6

Emotion's weight.

Emotion	Weight Value
Fear	0.3
Angry	0.29
Sad	0.45
Surprise	0.58
Happy	0.59
Neutral	0.8

5.1. Real-time web based attention system

The Flask framework is used to build a web application that gives learners the ability to monitor their cognitive state while they are engaged in online learning. The visual representation of the front end

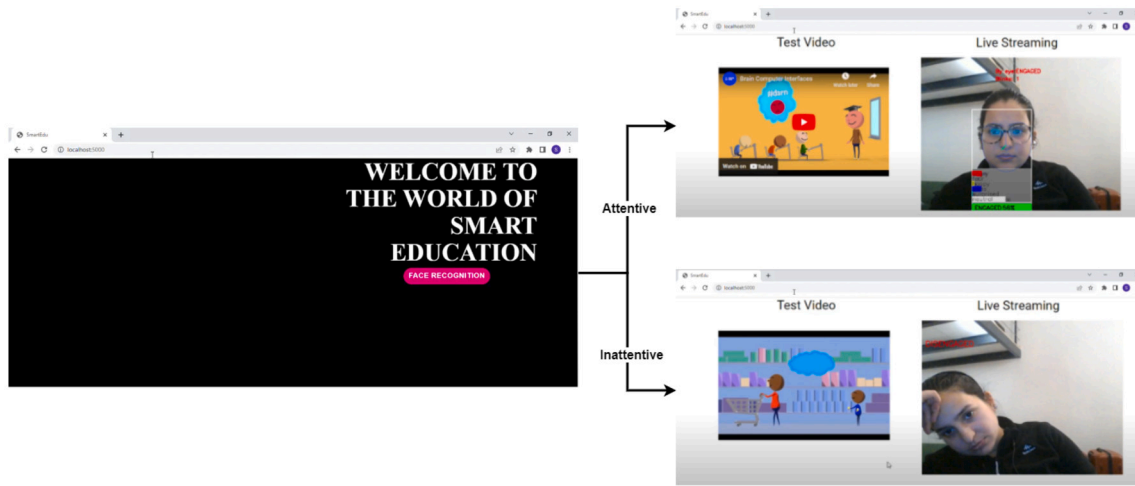


Fig. 12. Real-time working model for attention and inattention recognition based on OT-EDFA.

Algorithm 1 Facial emotion recognition algorithm.

Input : S_{ID} $\leftarrow$ unique student id
 δ_E $\leftarrow$ label for facial emotions
 $E \in \{\text{happy, sad, angry, neutral, surprised}\}$

Output: $WT_{Student}$ $\leftarrow$ Student Weights
 $WT_{Emotion}$ $\leftarrow$ Emotion Weights
 $MT_{emotion}$ $\leftarrow$ Emotion Weights Matrix
 $WT_{prediction}$ $\leftarrow$ prediction weights
 $\delta_{emotion}$ $\leftarrow$ Predicted Emotion

BEGIN:
Random Initialization $\rightarrow WT_{Student}$ and $WT_{Emotion}$
For $a = 1$ to $\text{len}(S_{ID})$
 For $b = 1$ to $\text{len}(\delta_E)$
 1. $Detect_face \leftarrow$ Haar cascade
 2. **CLASSIFY :** $Detect_face \rightarrow \delta_E$
 where $E \in \{\text{happy, sad, angry, neutral, surprised}\}$
 3. $\delta_{emotion} \leftarrow$ Apply Ensemble Learning model on $Detect_face$
 4. $WT_{prediction}[a][b] = \sum_{a=0}^{\text{len}(S_{ID})} \sum_{b=0}^{\text{len}(\delta)} WT_{Student} WT_{Emotion}$
 5. $S_{ID}(\delta_{emotion}) = MT_{emotion}^{10} \times WT_{prediction}$
OUTPUT: $\delta_{emotion}$ and $WT_{Emotion} \rightarrow$ recognised the emotional information of a student to help generate the AI (Attention index) for the Cognitive state prediction.
END

is shown in Fig. 12. The networks are implemented on VS code with Python 3.7 (Python Software), Scikit-Learn 0.21.5, Keras 2.3.3, TensorFlow 2.4.1 and many important deep learning libraries required for implementation. Flask 1.0.2 is used to develop the web application.

6. Experimental results

This section presents the outcomes of the experimental investigation of the proposed system based on the ensemble learning model. Four metrics will be used to compare the models: Precision, Sensitivity, F1 score, and Specificity. We will analyse the results based on these metrics to determine the performance of the models. To evaluate the system, fifty participants aged between 19 and 25, consisting of undergraduate and graduate students from diverse academic fields, were selected. The participants were asked to watch video lectures lasting for 2 minutes and 20 seconds on related and unrelated topics. The lectures comprised both verbal and diagrammatic information. Fig. 13 shows how analysing different facial features using computer vision helps determine the learner's cognitive state. Initially, in an online session, a video of the learner is captured, and images are extracted frame by frame. These images are obtained by using the built-in camera. Next, each frame of the collected images is processed using the Haar Cascade approach to detect the face region in each frame. After detecting the face, facial features are required to recognise the corresponding

Table 7

Test Comparison results between existing and ensemble models for Fer2013 dataset.

MODEL	Precision (%)	Sensitivity (%)	F1 score(%)	Specificity (%)
VGG19	79.05	79.64	79.40	79.73
ResNet50	80.21	80.33	80.57	80.97
FT-EDFA	88.19	88.28	88.56	88.78
FC-EDFA	88.12	88.47	88.69	88.33
OT-EDFA	89.66	89.59	89.82	89.91

emotion. To extract these facial features, facial landmark detection is performed using an approach shown in Fig. 14.

This approach identifies the key points on the face, which are then used to recognise the emotion being expressed. The resulting face images are then used as input to our proposed approaches, namely FT-EDFA, FC-EDFA, and OT-EDFA, which facilitate facial emotion recognition. To identify the cognitive state, Algorithm 1 is employed, which accurately predicts the cognitive state. The dataset is split 60%, 20%, and 20% amongst training, testing, and validation sets.

Fig. 15 depicts the overall process of the proposed approach. The original input image is first processed to extract facial landmarks, which are then used to extract facial features. These facial features are then fed into the ensemble learning model to classify the facial emotion. According to Fig. 15, the neutral emotion is recognised for the input image. Based on Algorithm 1, the cognitive state level is identified as attentive based on the calculation of AI value.

6.1. Comparative results of ensemble learning approach for various datasets

Three different ensemble-based approaches were proposed and implemented on four standard datasets, including FER-2013, RAF-DB, CK+, and an own collected dataset. The performance of the proposed approaches was compared using metrics such as precision, sensitivity, F1, and specificity. The results showed that the OT-EDFA approach consistently outperformed other approaches, including FC-EDFA, FT-EDFA, VGG19, and ResNet50, across all four datasets based on Table 7, Table 8, Table 9 and Table 10. Specifically, for Fer-2013, RAF-DB, CK+, and the own collected dataset, the OT-EDFA approach achieved the highest precision (89.66%, 90.26%, 92.26%, and 94.26%, respectively), sensitivity (89.59%, 90.59%, 92.59%, and 94.40%, respectively), F1 score (89.52%, 90.11%, 92.11%, and 94.51%, respectively), and specificity (89.91%, 90.79%, 92.79%, and 94.69%, respectively).

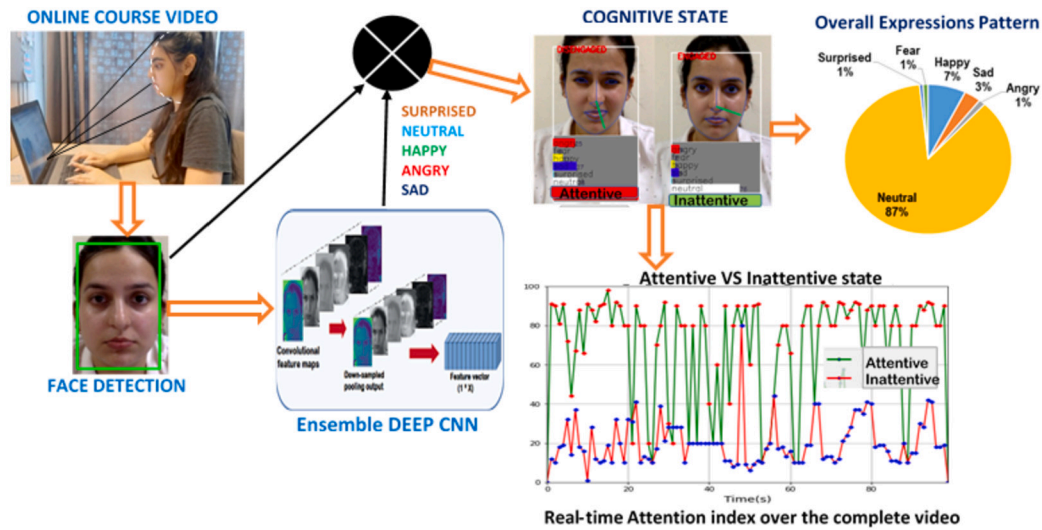


Fig. 13. Real-time working model for attention and inattention recognition based on OT-EDFA.

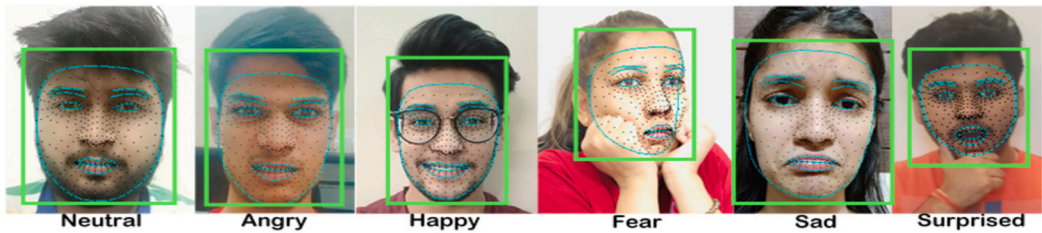


Fig. 14. Key-points representation of different emotions.

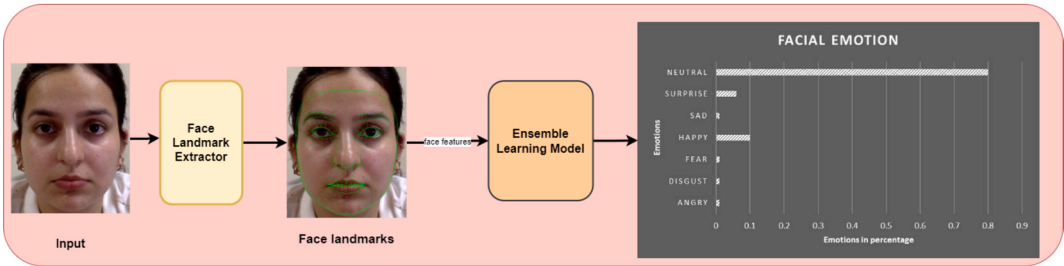


Fig. 15. Face Emotion recognition using ensemble learning model.

Table 8
Test Comparison results between existing and ensemble models for RAF DB dataset.

MODEL	Precision (%)	Sensitivity (%)	F1 score(%)	Specificity (%)
VGG19	84.53	84.34	84.54	84.66
ResNet50	85.20	85.13	85.74	85.45
FT-EDFA	89.19	89.68	89.56	89.81
FC-EDFA	89.02	89.17	89.26	89.23
OT-EDFA	90.26	90.59	90.11	90.79

Table 9
Test Comparison results between existing and ensemble models for CK+ dataset.

MODEL	Precision (%)	Sensitivity (%)	F1 score(%)	Specificity (%)
VGG19	86.53	86.14	86.46	86.45
ResNet50	87.40	87.13	87.34	87.50
FT-EDFA	90.09	90.18	90.22	90.29
FC-EDFA	90.13	90.15	90.36	90.39
OT-EDFA	92.26	92.59	92.11	92.79

Table 10
Test Comparison results between existing and ensemble models for OWN dataset.

MODEL	Precision (%)	Sensitivity (%)	F1 score(%)	Specificity (%)
VGG19	88.53	88.14	88.46	88.48
ResNet50	88.65	88.23	88.34	88.58
FT-EDFA	91.09	91.18	91.22	91.70
FC-EDFA	92.33	92.45	92.36	92.89
OT-EDFA	94.26	94.40	94.51	94.69

6.2. Confusion matrix results for ensemble learning approaches

The confusion matrix offers valuable information regarding the accuracy of predicted labels compared to actual labels. It divides the results into two categories: attentive and inattentive. Based on the results of the study, the proposed OT-EDFA method demonstrated a high level of accuracy, correctly classifying 436 out of 450 facial emotion images. The FC-EDFA model had the second-highest accuracy rate, with 431 out of 450 images correctly classified, followed closely by the

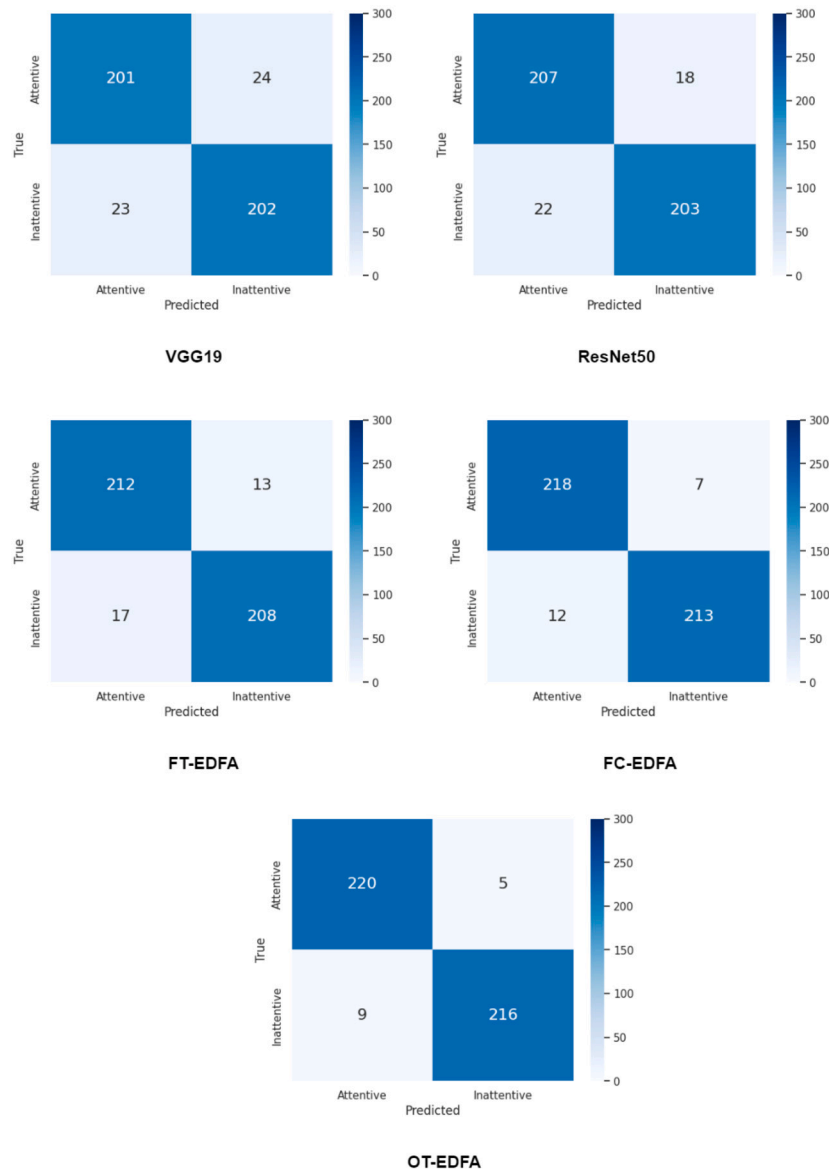


Fig. 16. Confusion matrix using ensemble learning model.

FT-EDFA model, which accurately identified 420 facial images. However, in the existing systems, VGG19 and ResNet50 had lower accuracy rates, correctly classifying only 403 and 410 out of 450 images, respectively. These findings suggest that the ensemble models outperformed the existing systems in terms of accuracy and reliability. By referring to Fig. 16, it becomes evident that the OT-EDFA approach outperforms other ensemble and existing methods in terms of efficiency.

6.3. Accuracy-based results for ensemble learning approaches

The results presented in Fig. 17 revealed that the OT-EDFA method exhibited the highest accuracy rates for both training and testing, at 93.11% and 92.67%, respectively. The FC-EDFA and FT-EDFA models achieved lower accuracy rates, with 92.34% and 91.12% training accuracy and 92.08% and 91.01% testing accuracy, respectively. The VGG19 and ResNet50 models had even lower accuracy rates, with 88.91% and 88.32% training accuracy and 87.12% and 86.89% testing accuracy, respectively. These results indicate that the OT-EDFA approach demonstrated superior accuracy and efficiency compared to the other models.

The graphical representation in Fig. 18 displays the accuracy rates of different models on various datasets. The OT-EDFA method demon-

strated excellent performance on all datasets, achieving accuracy rates of 78.5% (FER-2013), 82.23% (CK+), 79.55% (RAB-DF), and 93.11% (own dataset) compared to other approaches. These results suggest the OT-EDFA method is highly effective and reliable for emotion recognition tasks on various datasets.

6.4. Results for false discovery rate

The false discovery rate (FDR) refers to the proportion or percentage of false discoveries among all the discoveries made in a study or analysis. For instance, if a researcher claims to have made 100 discoveries, but 10 of them are false, then the FDR would be 10%, indicating that 10% of the discoveries are actually false positives. In other words, FDR measures the rate of erroneous rejections of the null hypothesis in multiple-hypothesis testing. According to Table 11, the false discovery rate (FDR) of existing models and all proposed ensemble-based models was compared. It was observed that the OT-EDFA approach achieved the lowest FDR compared to the other models.

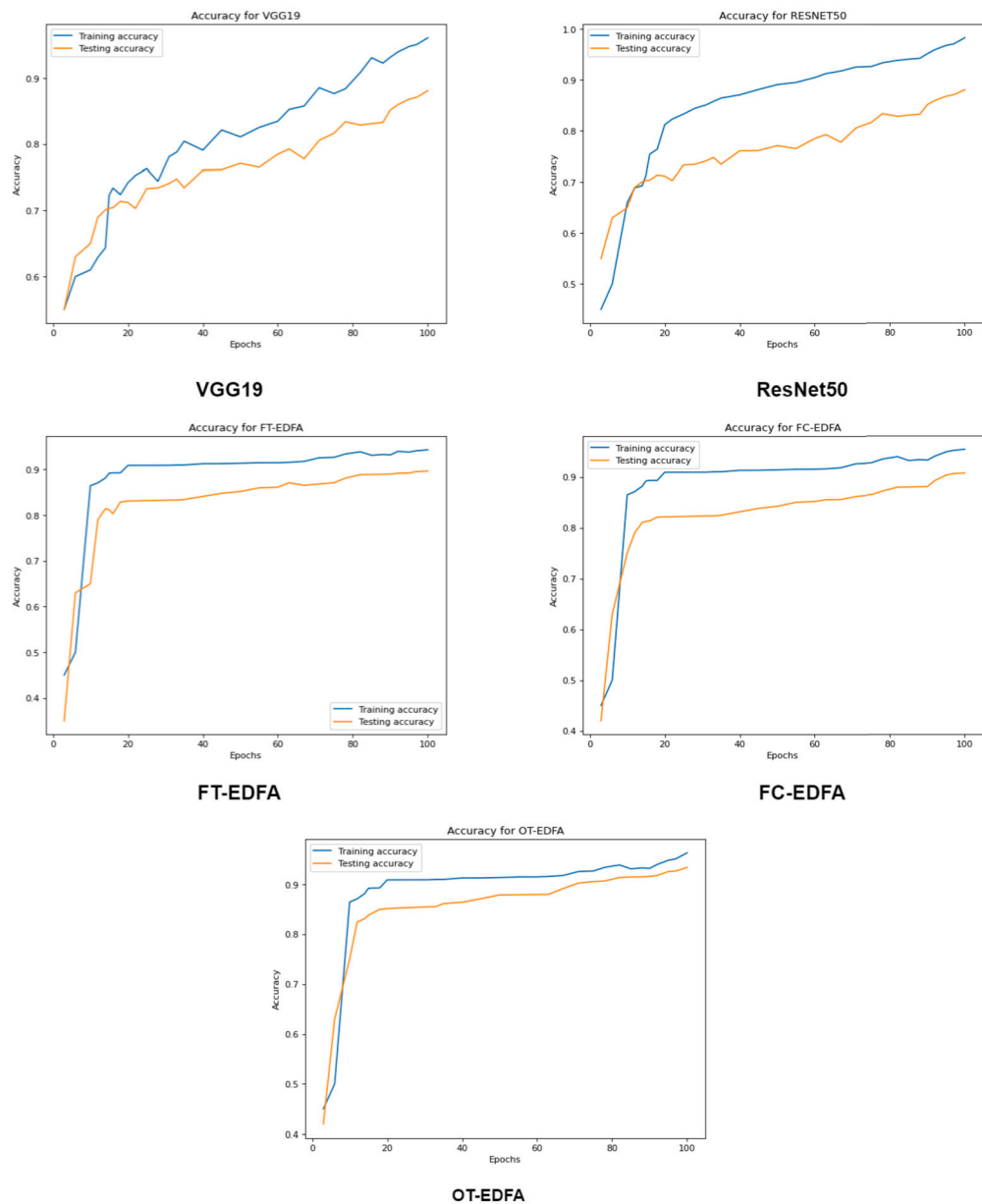


Fig. 17. Accuracy using ensemble learning model.

Table 11
False discovery rate (FDR) results between existing and ensemble models for OWN dataset.

MODEL	FDR
VGG19	0.0661
ResNet50	0.0611
FT-EDFA	0.0548
FC-EDFA	0.0462
OT-EDFA	0.0125

7. Temporal facial emotion recognition for cognitive states

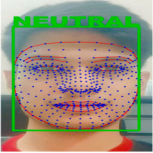
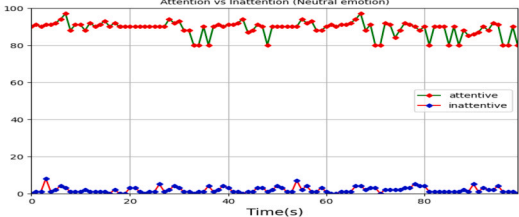
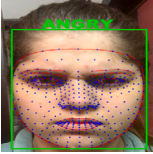
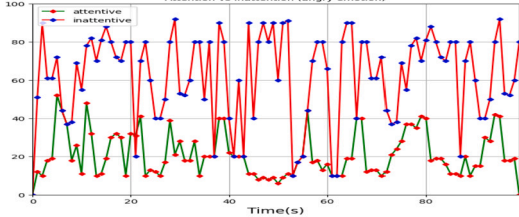
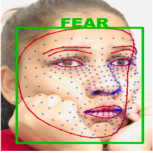
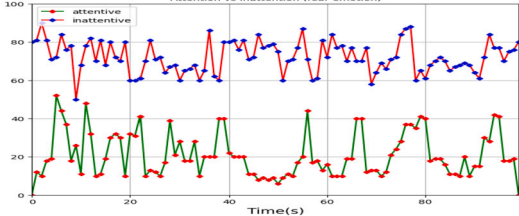
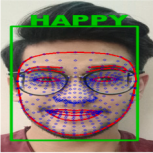
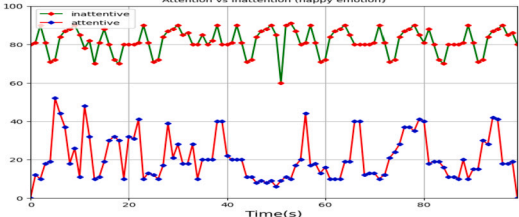
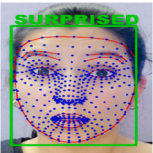
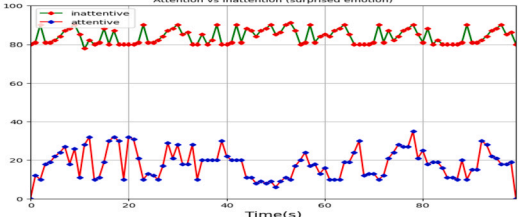
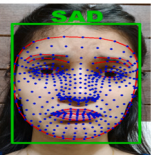
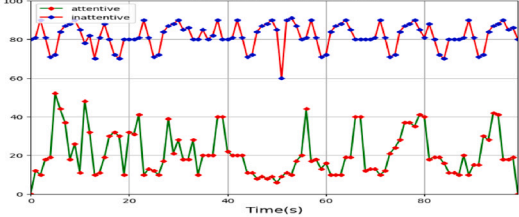
This section focuses on real-time facial emotion recognition to analyse the cognitive state of students. Images extracted from real-time video streaming are analysed to identify six facial emotions such as neutral, happy, surprised, fear, sad, and angry. The temporal graphs

corresponding to those emotions are presented in Table 12 to represent cognitive states. It is evident from the graphs that if the emotion is neutral, happy, or surprised, the student’s attentiveness is high. In contrast, if the emotion is fear, sad, or angry, the learner’s attentiveness is very low, and the inattentive cognitive state is high.

8. Discussion

E-Learning offers the convenience of attending lectures from home and watching recorded lectures, but many students fail to complete online courses. This is due to several factors, including the lack of physical and student-to-student interaction. A cognitive state assessment is necessary to enhance e-learning environments and provide interactive feedback. We propose an ensemble learning-based framework for predicting student attention to address this. Ensemble learning is a powerful approach to facial emotion detection that combines the predictions of multiple models to achieve higher accuracy and better performance. In the proposed ensemble model, we combine the important characteristic of VGG19 and ResNet50. We have implemented and evaluated

Table 12
Temporal representation of attentive or inattentive based on facial emotion.

Emotions	Emotion Detected	Temporal Emotion graph	Cognitive State
	Neutral		Attentive
	Angry		Inattentive
	Fear		Inattentive
	Happy		Attentive
	Surprised		Attentive
	Sad		Inattentive

the statistical performance of VGG19, ResNet50 and ensemble models on various datasets such as FER-2013, ResNet50, RAF-DB and our own created datasets. Ensemble models gave better results than VGG19 and ResNet50. The proposed algorithm calculated the AI value, and based on that value, a decision about the cognitive state is made. The present work is compared with other existing systems to evaluate the efficiency of our proposed approach.

Our proposed model stands out from previous studies in several ways. Firstly, unlike many studies that suffer from imbalanced datasets, our model is trained on a balanced dataset that includes benchmarked and own collected datasets. Secondly, we took advantage of pre-trained models to improve the efficiency of the training process. Finally, we proposed three ensemble models for real-time facial emotion recognition and summarised their findings in the section above. The results of

Table 13
Comparison of the existing models with a proposed system for Real-time Cognitive State Assessment.

S.NO.	AUTHOR	DATASET USED	Model	ACCURACY (in %)
1.	Jun et al. (2018)	FER-2013	VGG16	73.06
2.	Gera and Balasubramanian (2021)	VGG19	SCAN	65.14
3.	Li and Lima (2021)	FER-2013+CK(+)	ResNet50	82.9
4.	Tong et al. (2022)	RAFD-B	AWOBNNet	83.6
5.	Proposed Model	FER-2013+ CK(+) + RAF-DB+ OWN Dataset	OT-EDFA FC-EDFA FT-EDFA	93.11 92.34 91.12

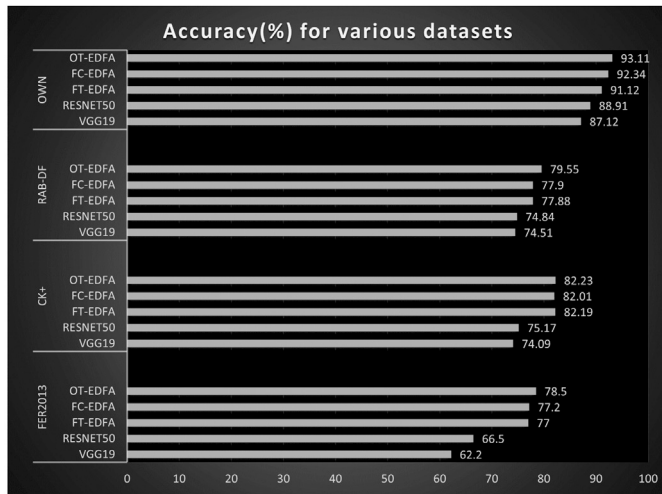


Fig. 18. Graphical representation of overall accuracy evaluation in percentage over different datasets.

the proposed ensemble models shown in the below table are evaluated on own created dataset. The results of each model are summarised in the following Table 13. Our proposed ensemble model outperformed existing systems for cognitive detection in real-time.

9. Conclusion and future scope

Making e-learning effective is crucial in today's digital age, where online education has become more prevalent than ever before. To make e-learning effective, it is essential to ensure that providing learners with regular feedback and support can help them stay motivated and on track with their learning goals. Using computer vision techniques to monitor the cognitive state of the learner has gained a lot of attention from researchers around the world. Research studies have shown exciting results while utilising different neural networks. Our study aims to help educators monitor learner's behaviour in adaptive learning platforms and provide real-time pedagogical support based on their performance. One way to understand whether learners are attentive is to observe their facial expressions while studying. Therefore, our proposed approach involves capturing the facial features of learners in online learning environments. Various deep-learning models have been developed and deployed for facial emotion recognition. This study applied transfer learning to VGG19 and ResNet50 deep CNN models for generating ensemble deep CNN models. By applying transfer learning to both VGG19 and ResNet50, the ensemble model can benefit from the strengths of both models, resulting in improved performance in recognising facial emotions. The three ensemble models are generated by combining the features of VGG19 and ResNet50: FT-EDFA, FC-EDFA and OT-EDFA. The datasets used to train and test the models are FER-2013, CK+, RAF-DB and the own created dataset. An algorithm is proposed for classifying the emotion in real-time based on the ensemble model. The input

image is captured using a device camera. The face area is obtained by implementing the Haar cascade algorithm on the input image. Haar cascade is used because it gives efficient results in real-time scenarios. Face features are extracted using the landmark approach for better facial emotion classification. Then, the facial feature map is fed to the ensemble model to recognise the facial emotion. The emotions on the face are detected by analysing the expression of the face by investigating the facial points (features). The value of AI (Attention Index) is calculated using the proposed algorithm. Therefore, the decision about the cognitive state, either attentive or inattentive, is given. The experimental results showed that the ensemble model- OT-EDFA, outperformed other models and achieved an accuracy of 93.11%. Moreover, we have developed a web application to easily deploy the proposed ensemble-based cognitive detection system. The proposed system significantly improves facial emotion recognition performance in a real-time learning environment.

In future work, combining the proposed approach with EEG-based approaches could provide a more comprehensive understanding of learners' cognitive states. Investigating the impact of additional features such as audio, eye gaze, and body posture in conjunction with facial features could improve the accuracy and reliability of the proposed cognitive detection system. Evaluating the proposed system with a larger dataset and incorporating more diverse emotional expressions could improve the system's ability to generalise to different learners and learning environments. These are all important directions for future research in this area. Moreover, the proposed system could be extended to other domains, such as healthcare, where it could monitor patient's cognitive state during medical procedures.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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